The CONTENTS Procedure

Data Set Name	WORK.COUPONS	**Observations**	12684
Member Type	DATA	**Variables**	26
Engine	V9	**Indexes**	0
Created	02/05/2026 15:54:17	**Observation Length**	224
Last Modified	02/05/2026 15:54:17	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	22
First Data Page	1
Max Obs per Page	584
Obs in First Data Page	561
Number of Data Set Repairs	0
Filename	/saswork/SAS_work3CBA0001009A_odaws01-usw2.oda.sas.com/SAS_work512F0001009A_odaws01-usw2.oda.sas.com/coupons.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	3914143
Access Permission	rw-r--r--
Owner Name	u64508924
File Size	3MB
File Size (bytes)	3014656

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Informat
16	Bar	Char	5	$5.	$5.
18	CarryAway	Char	1	$1.	$1.
17	CoffeeHouse	Char	5	$5.	$5.
20	Restaurant20To50	Char	3	$3.	$3.
19	RestaurantLessThan20	Char	3	$3.	$3.
26	Y	Num	8	BEST12.	BEST32.
9	age	Num	8	BEST12.	BEST32.
15	car	Char	1	$1.	$1.
6	coupon	Char	21	$21.	$21.
1	destination	Char	15	$15.	$15.
25	direction_opp	Num	8	BEST12.	BEST32.
24	direction_same	Num	8	BEST12.	BEST32.
12	education	Char	24	$24.	$24.
7	expiration	Char	2	$2.	$2.
8	gender	Char	6	$6.	$6.
11	has_children	Num	8	BEST12.	BEST32.
14	income	Char	15	$15.	$15.
10	maritalStatus	Char	17	$17.	$17.
13	occupation	Char	10	$10.	$10.
2	passanger	Char	9	$9.	$9.
4	temperature	Num	8	BEST12.	BEST32.
5	time	Char	4	$4.	$4.
22	toCoupon_GEQ15min	Num	8	BEST12.	BEST32.
23	toCoupon_GEQ25min	Num	8	BEST12.	BEST32.
21	toCoupon_GEQ5min	Num	8	BEST12.	BEST32.
3	weather	Char	5	$5.	$5.

Descriptive Statistics for Numeric Variables

The MEANS Procedure

Variable	N	N Miss	Mean	Median	Std Dev	Minimum	Maximum
temperature	12684	0	63.3017975	80.0000000	19.1544858	30.0000000	80.0000000
age	10349	2335	29.8878152	26.0000000	7.6972751	21.0000000	46.0000000
has_children	12684	0	0.4141438	0	0.4925930	0	1.0000000
toCoupon_GEQ5min	12684	0	1.0000000	1.0000000	0	1.0000000	1.0000000
toCoupon_GEQ15min	12684	0	0.5614948	1.0000000	0.4962235	0	1.0000000
toCoupon_GEQ25min	12684	0	0.1191265	0	0.3239500	0	1.0000000
direction_same	12684	0	0.2147588	0	0.4106711	0	1.0000000
direction_opp	12684	0	0.7852412	1.0000000	0.4106711	0	1.0000000
Y	12684	0	0.5684327	1.0000000	0.4953144	0	1.0000000

Frequency Distributions of Categorical Variables

The FREQ Procedure

destination	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Home	3237	25.52	3237	25.52
No Urgent Place	6283	49.53	9520	75.06
Work	3164	24.94	12684	100.00

passanger	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Alone	7305	57.59	7305	57.59
Friend(s)	3298	26.00	10603	83.59
Kid(s)	1006	7.93	11609	91.52
Partner	1075	8.48	12684	100.00

weather	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Rainy	1210	9.54	1210	9.54
Snowy	1405	11.08	2615	20.62
Sunny	10069	79.38	12684	100.00

time	Frequency	Percent	Cumulative Frequency	Cumulative Percent
10AM	2275	17.94	2275	17.94
10PM	2006	15.82	4281	33.75
2PM	2009	15.84	6290	49.59
6PM	3230	25.47	9520	75.06
7AM	3164	24.94	12684	100.00

coupon	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Bar	2017	15.90	2017	15.90
Carry out & Take away	2393	18.87	4410	34.77
Coffee House	3996	31.50	8406	66.27
Restaurant(20-50)	1492	11.76	9898	78.04
Restaurant(<20)	2786	21.96	12684	100.00

expiration	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1d	7091	55.91	7091	55.91
2h	5593	44.09	12684	100.00

gender	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Female	6511	51.33	6511	51.33
Male	6173	48.67	12684	100.00

maritalStatus	Frequency	Percent	Cumulative Frequency	Cumulative Percent

maritalStatus	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Divorced	516	4.07	516	4.07
Married partner	5100	40.21	5616	44.28
Single	4752	37.46	10368	81.74
Unmarried partner	2186	17.23	12554	98.98
Widowed	130	1.02	12684	100.00

education	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Associates degree	1153	9.09	1153	9.09
Bachelors degree	4335	34.18	5488	43.27
Graduate degree (Masters	1852	14.60	7340	57.87
High School Graduate	905	7.13	8245	65.00
Some High School	88	0.69	8333	65.70
Some college - no degree	4351	34.30	12684	100.00

occupation	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Architectu	175	1.38	175	1.38
Arts Desig	629	4.96	804	6.34
Building &	44	0.35	848	6.69
Business &	544	4.29	1392	10.97
Community	241	1.90	1633	12.87
Computer &	1408	11.10	3041	23.98
Constructi	154	1.21	3195	25.19
Education&	943	7.43	4138	32.62
Farming Fi	43	0.34	4181	32.96
Food Prepa	298	2.35	4479	35.31
Healthcare	486	3.83	4965	39.14
Installati	133	1.05	5098	40.19
Legal	219	1.73	5317	41.92
Life Physi	170	1.34	5487	43.26
Management	838	6.61	6325	49.87
Office & A	639	5.04	6964	54.90
Personal C	175	1.38	7139	56.28
Production	110	0.87	7249	57.15
Protective	175	1.38	7424	58.53
Retired	495	3.90	7919	62.43
Sales & Re	1093	8.62	9012	71.05
Student	1584	12.49	10596	83.54
Transporta	218	1.72	10814	85.26
Unemployed	1870	14.74	12684	100.00

income	Frequency	Percent	Cumulative Frequency	Cumulative Percent
$100000 or More	1736	13.69	1736	13.69
$12500 - $24999	1831	14.44	3567	28.12
$25000 - $37499	2013	15.87	5580	43.99
$37500 - $49999	1805	14.23	7385	58.22
$50000 - $62499	1659	13.08	9044	71.30
$62500 - $74999	846	6.67	9890	77.97
$75000 - $87499	857	6.76	10747	84.73
$87500 - $99999	895	7.06	11642	91.78
Less than $1250	1042	8.22	12684	100.00

car	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	12576	99.15	12576	99.15
C	21	0.17	12597	99.31
M	22	0.17	12619	99.49

car	Frequency	Percent	Cumulative Frequency	Cumulative Percent
S	22	0.17	12641	99.66
c	21	0.17	12662	99.83
d	22	0.17	12684	100.00

Bar	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	107	0.84	107	0.84
1~3	2473	19.50	2580	20.34
4~8	1076	8.48	3656	28.82
gt8	349	2.75	4005	31.58
less1	3482	27.45	7487	59.03
never	5197	40.97	12684	100.00

CoffeeHouse	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	217	1.71	217	1.71
1~3	3225	25.43	3442	27.14
4~8	1784	14.06	5226	41.20
gt8	1111	8.76	6337	49.96
less1	3385	26.69	9722	76.65
never	2962	23.35	12684	100.00

CarryAway	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	151	1.19	151	1.19
1	4672	36.83	4823	38.02
4	4258	33.57	9081	71.59
g	1594	12.57	10675	84.16
l	1856	14.63	12531	98.79
n	153	1.21	12684	100.00

RestaurantLessThan20	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	130	1.02	130	1.02
1~3	5376	42.38	5506	43.41
4~8	3580	28.22	9086	71.63
gt8	1285	10.13	10371	81.76
les	2093	16.50	12464	98.27
nev	220	1.73	12684	100.00

Restaurant20To50	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	189	1.49	189	1.49
1~3	3290	25.94	3479	27.43
4~8	728	5.74	4207	33.17
gt8	264	2.08	4471	35.25
les	6077	47.91	10548	83.16
nev	2136	16.84	12684	100.00

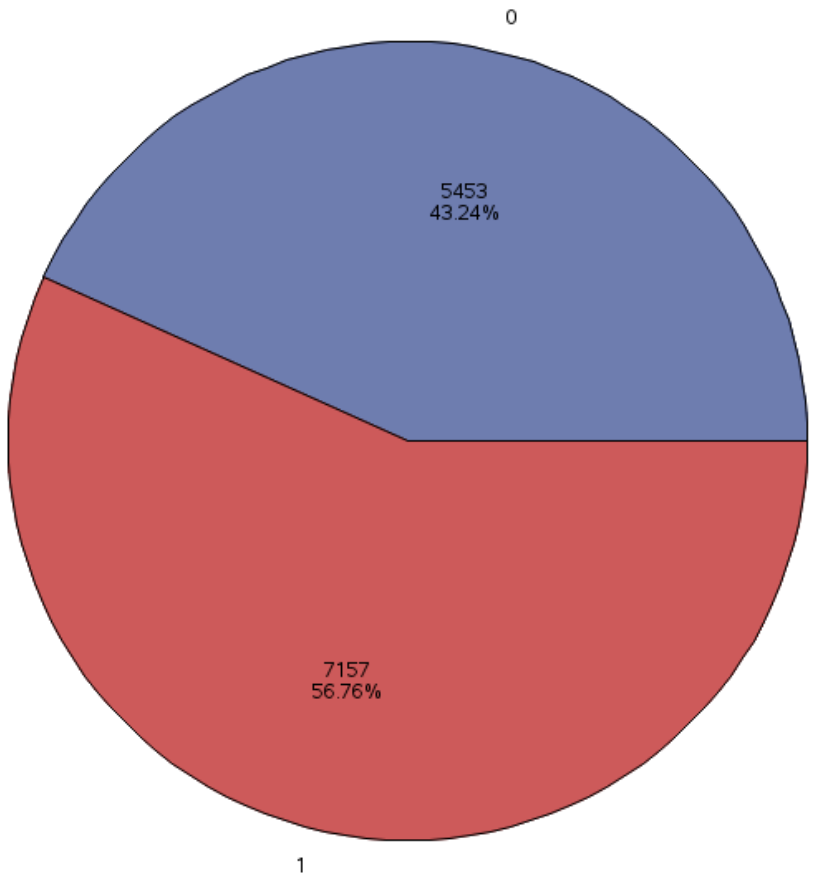
Missing Values Count

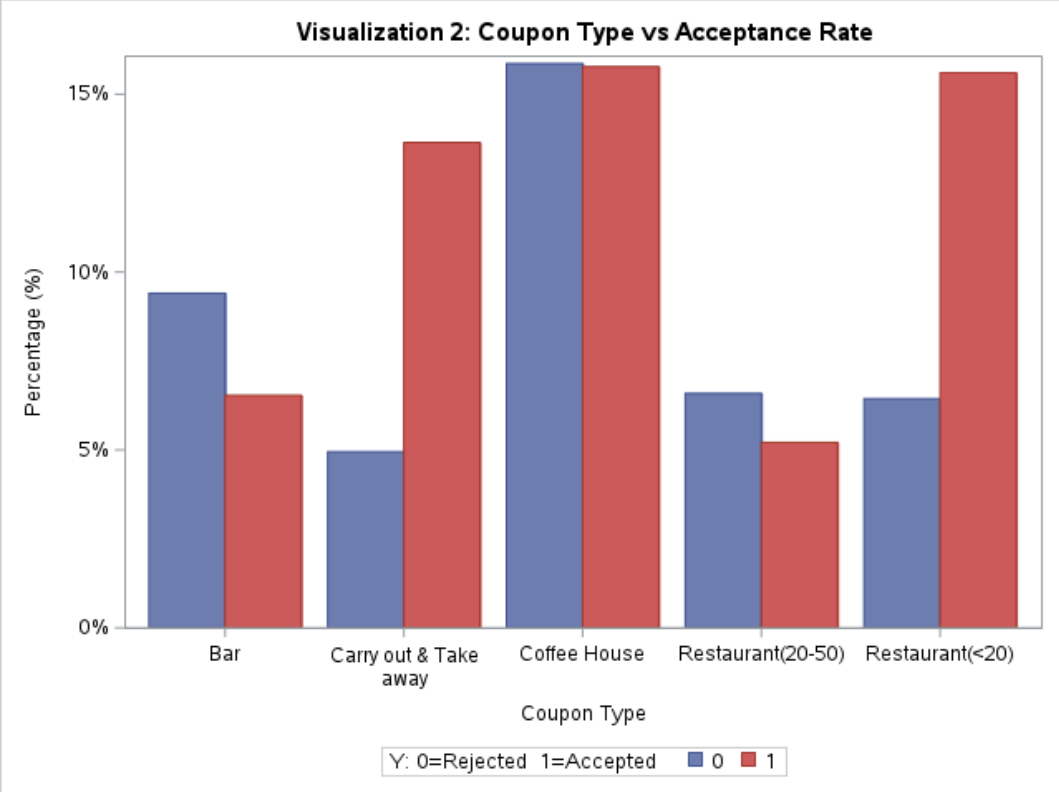
missing_age	missing_car	missing_Bar	missing_CoffeeHouse	missing_CarryAway	missing_RestaurantLessThan20	missing_Restaurant20To50
2335	12576	107	217	151	130	189

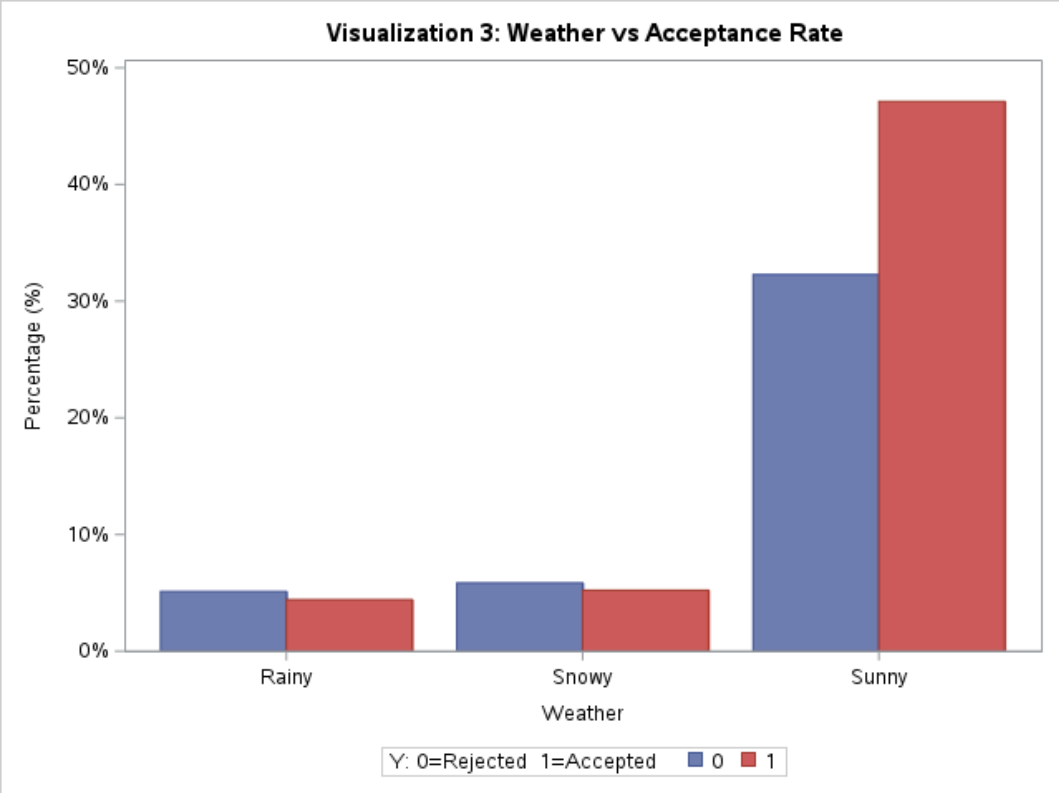
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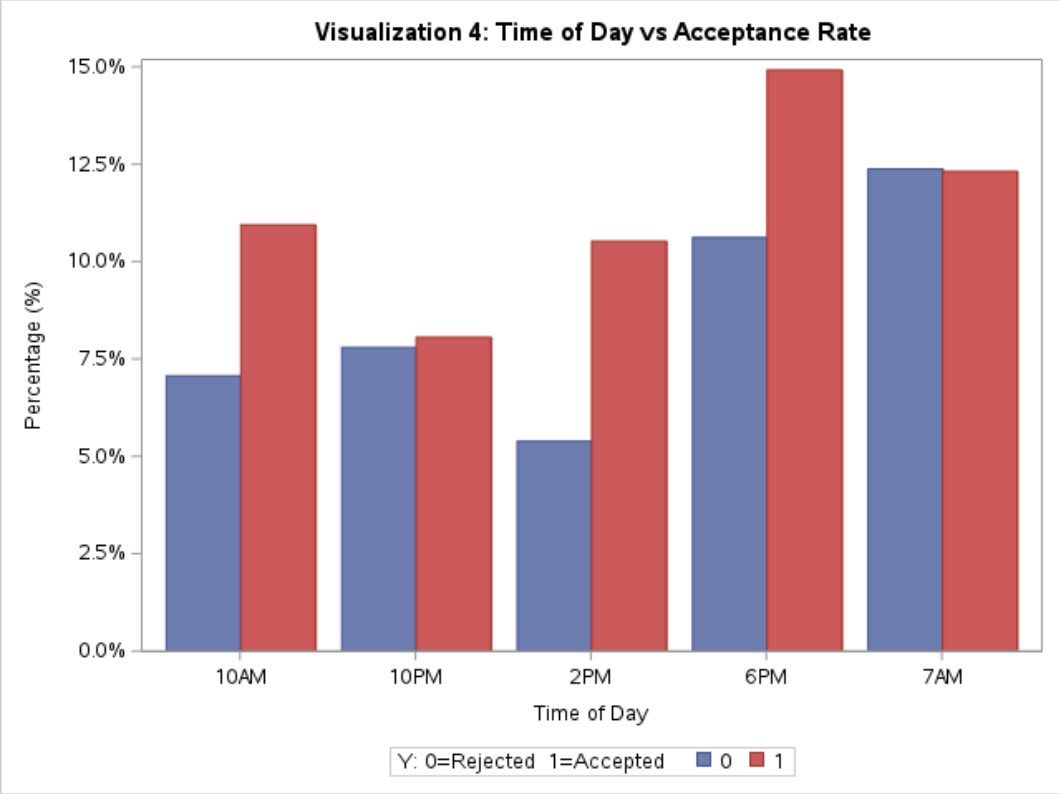
original_count	after_dedup_count	duplicate_count
12684	12610	74

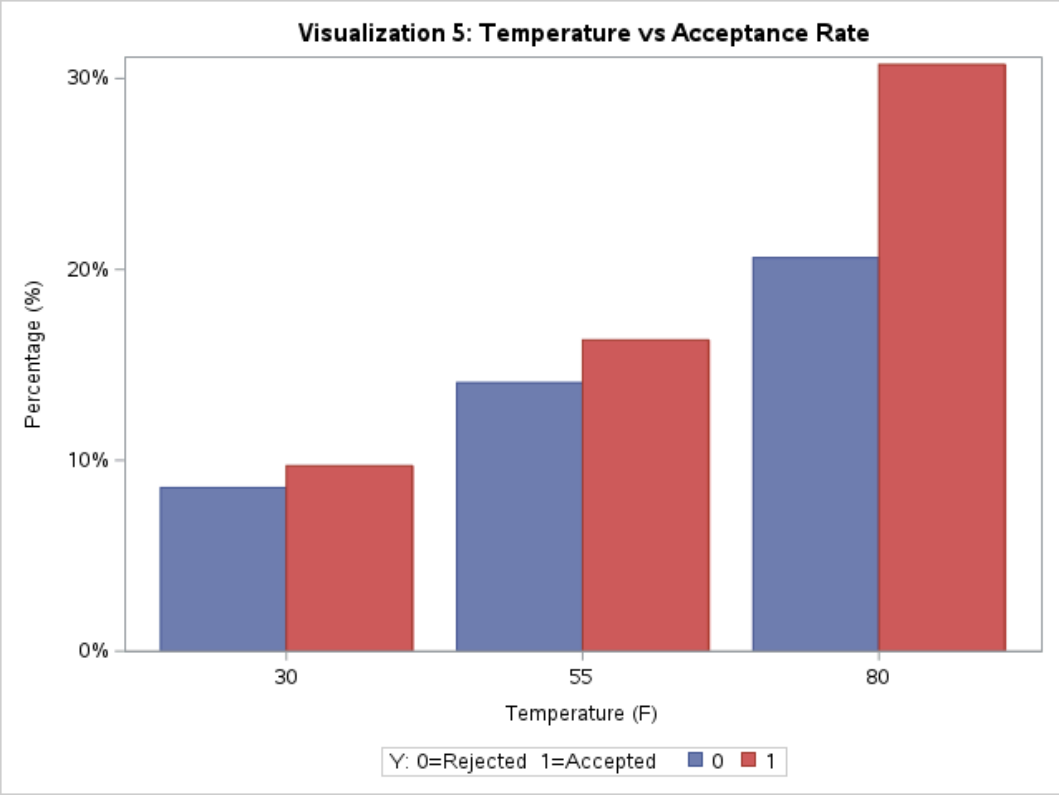
Visualization 1: Coupon Acceptance Rate (Y=1 Accepted, Y=0 Rejected)

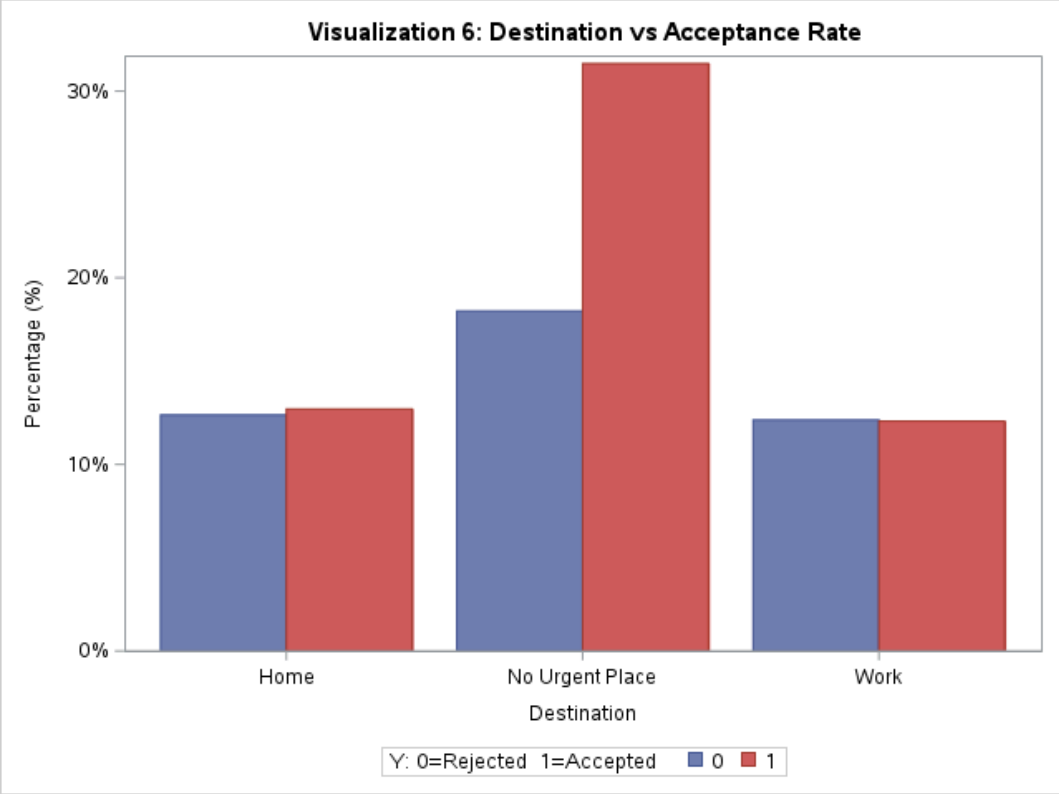


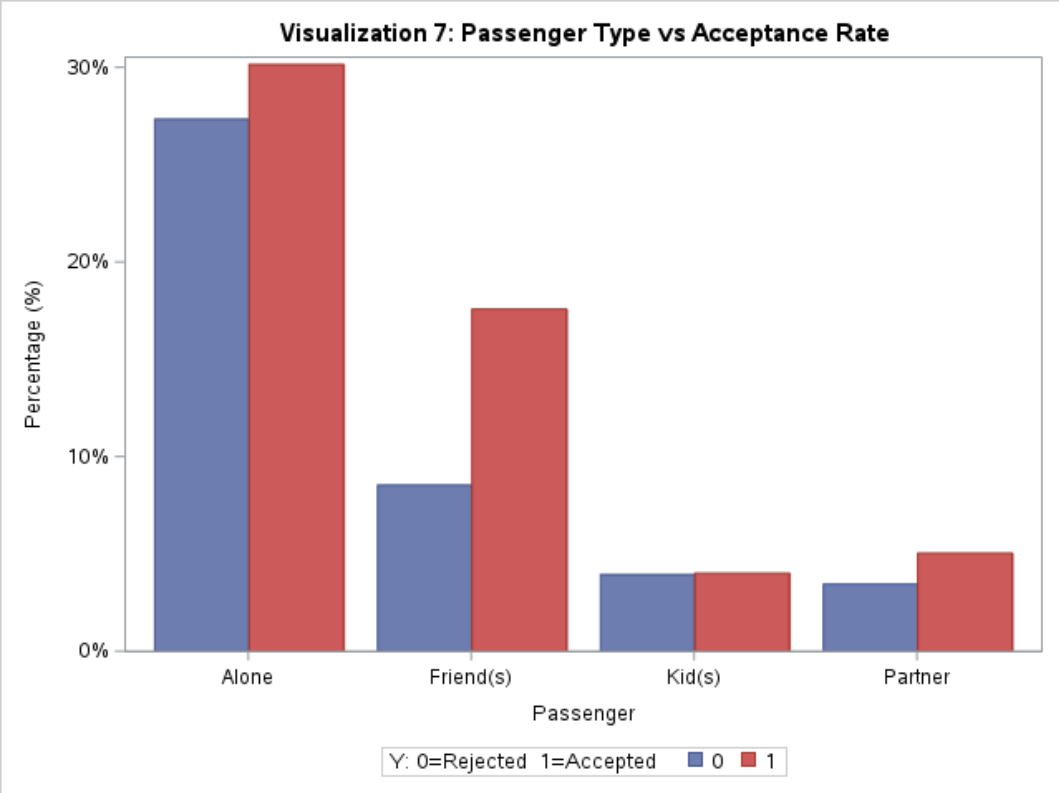

Visualization 2: Coupon Type vs Acceptance Rate
15%
10%
5%
0%
Percentage (%)
Bar
Carry out & Take away
Coffee House
Restaurant(20-50)
Restaurant(<20)
Coupon Type
Y: 0=Rejected 1=Accepted
0
1

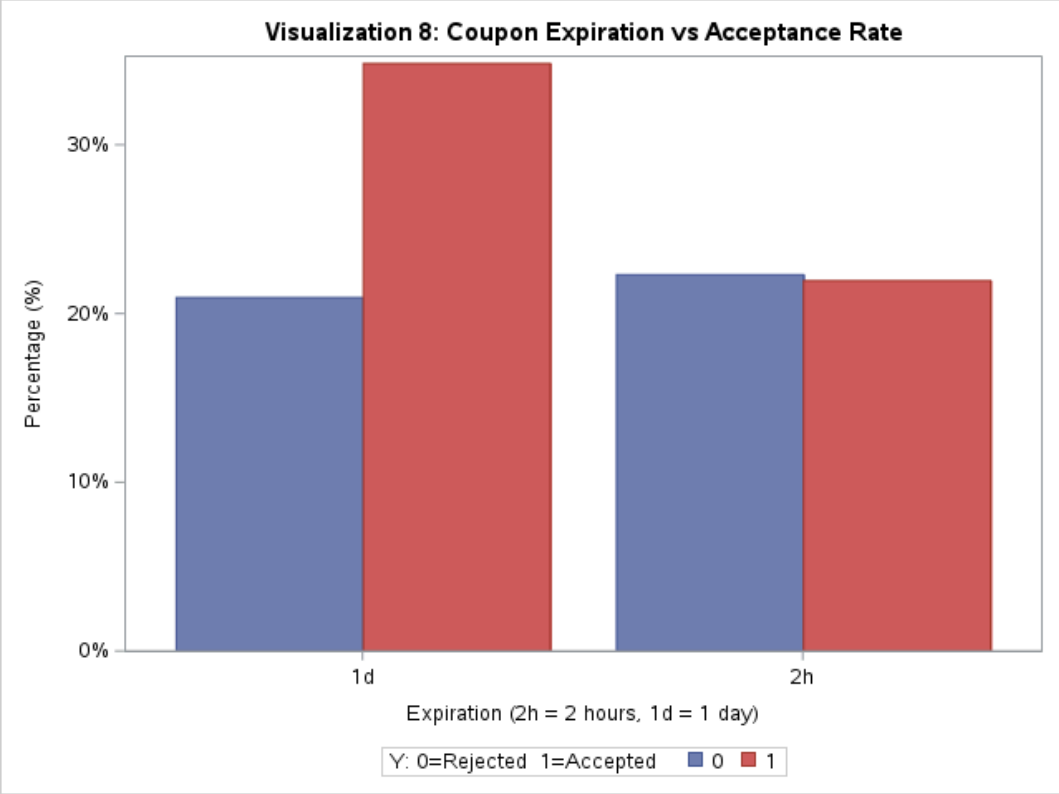

Visualization 3: Weather vs Acceptance Rate
50%
40%
30%
20%
10%
0%
Percentage (%)
Rainy
Snowy
Sunny
Weather
Y: 0=Rejected 1=Accepted
0
1

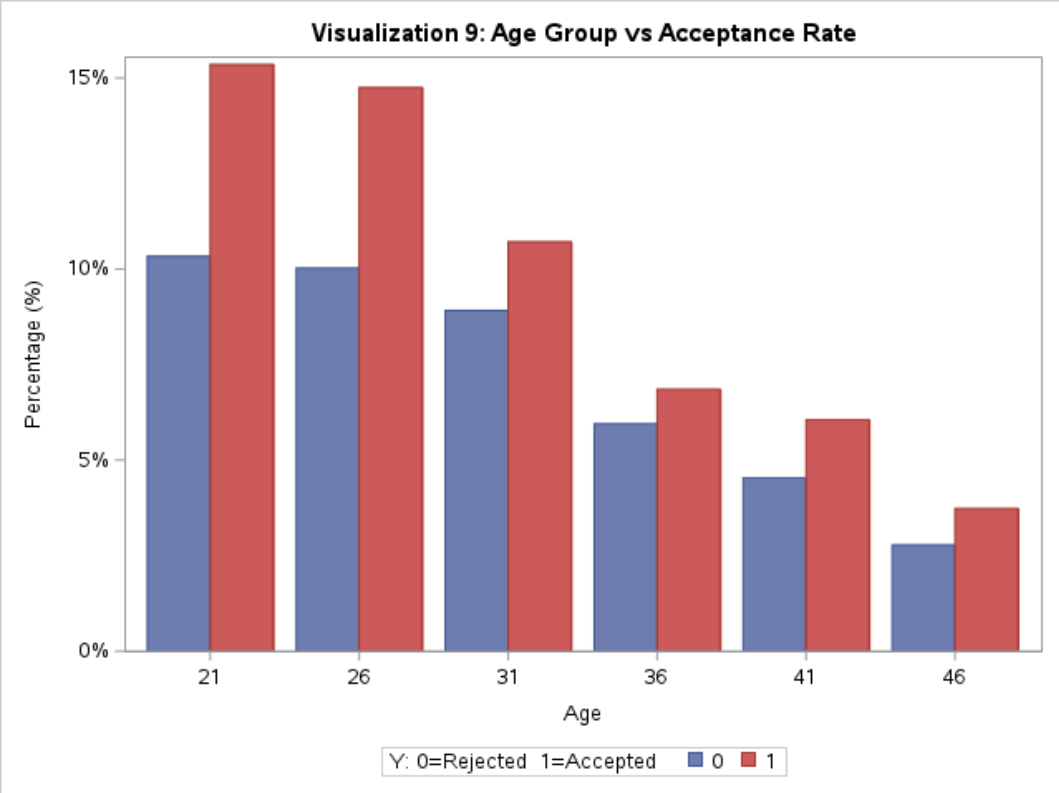

Visualization 4: Time of Day vs Acceptance Rate
15.0%
12.5%
10.0%
7.5%
5.0%
2.5%
0.0%
Percentage (%)
10AM
10PM
2PM
6PM
7AM
Time of Day
Y: 0=Rejected 1=Accepted
0
1

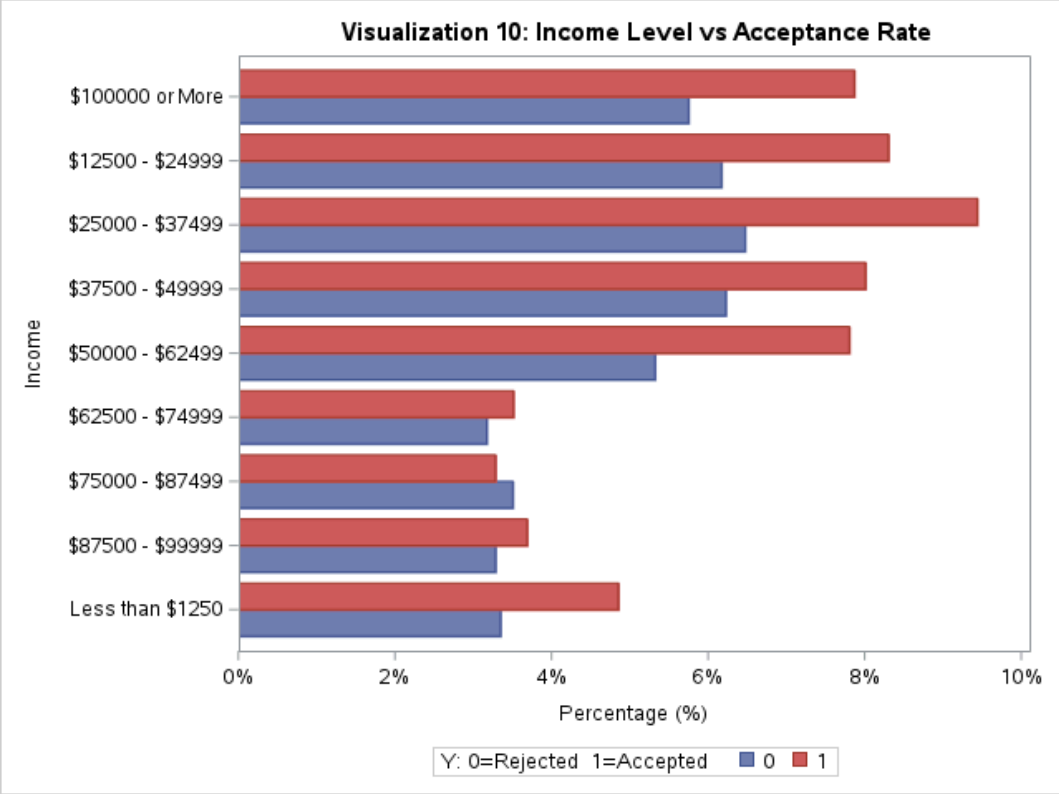

Visualization 5: Temperature vs Acceptance Rate
30%
20%
10%
0%
Percentage (%)
30
55
80
Temperature (F)
Y: 0=Rejected 1=Accepted
0
1

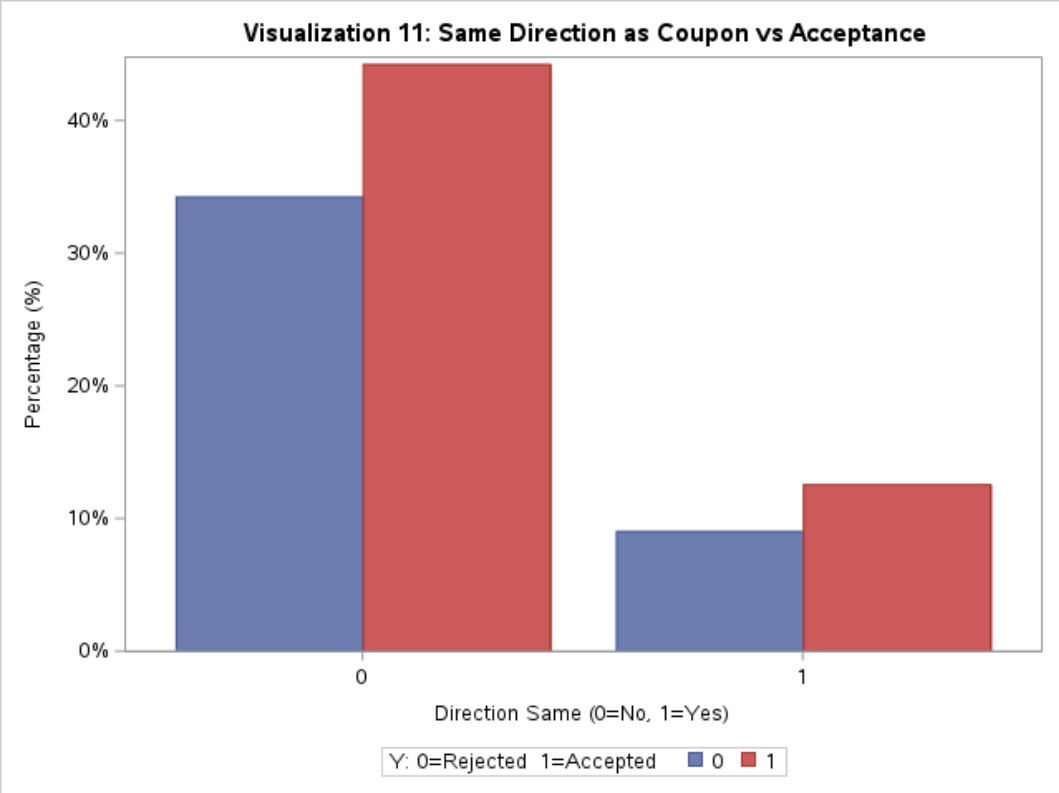

Visualization 6: Destination vs Acceptance Rate
30%
20%
10%
0%
Percentage (%)
Home
No Urgent Place
Work
Destination
Y: 0=Rejected 1=Accepted
0
1

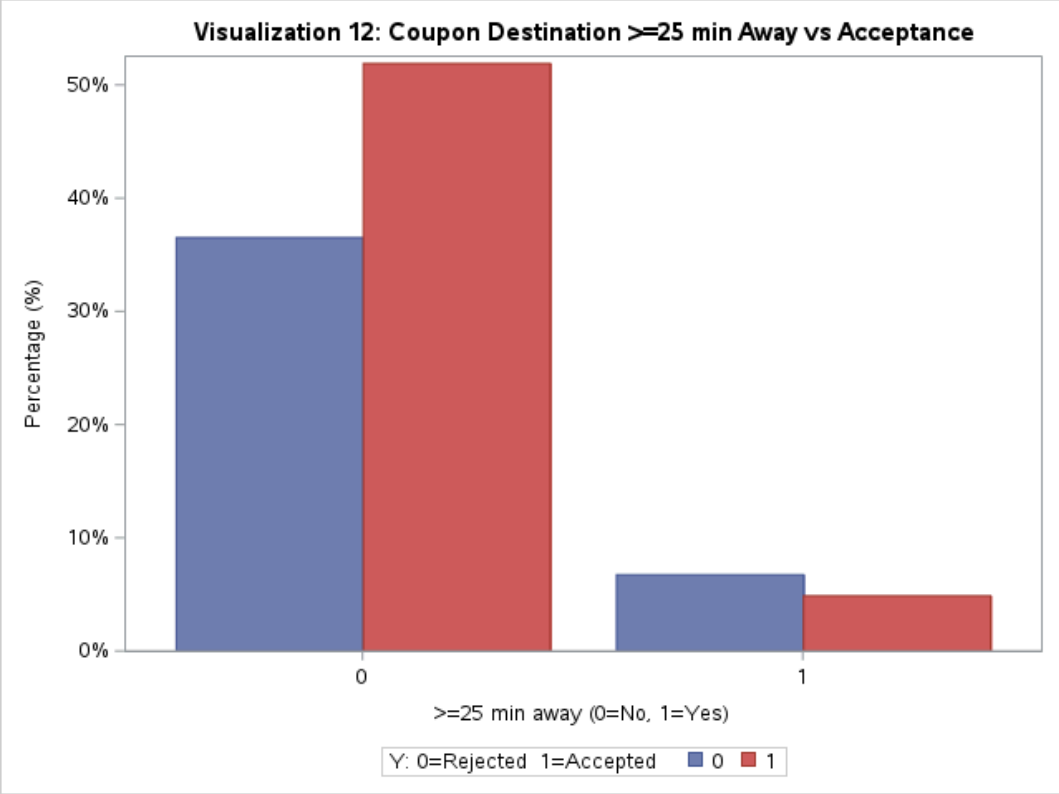

Visualization 7: Passenger Type vs Acceptance Rate
30%
20%
10%
0%
Percentage (%)
Alone
Friend(s)
Kid(s)
Partner
Passenger
Y: 0=Rejected 1=Accepted
0
1


Visualization 8: Coupon Expiration vs Acceptance Rate
30%
20%
10%
0%
Percentage (%)
1d
2h
Expiration (2h = 2 hours, 1d = 1 day)
Y: 0=Rejected 1=Accepted
0
1


Visualization 9: Age Group vs Acceptance Rate
15%
10%
5%
0%
Percentage (%)
21
26
31
36
41
46
Age
Y: 0=Rejected 1=Accepted
0
1


Visualization 10: Income Level vs Acceptance Rate
$100000 or More
$12500 - $24999
$25000 - $37499
$37500 - $49999
$50000 - $62499
$62500 - $74999
$75000 - $87499
$87500 - $99999
Less than $1250
Income
0%
2%
4%
6%
8%
10%
Percentage (%)
Y: 0=Rejected 1=Accepted
0
1


Visualization 11: Same Direction as Coupon vs Acceptance
40%
30%
20%
10%
0%
Percentage (%)
0
1
Direction Same (0=No, 1=Yes)
Y: 0=Rejected 1=Accepted
0
1



Pearson Correlation Matrix of Numeric Variables

The CORR Procedure

9 Variables:	temperature age has_children toCoupon_GEQ5min toCoupon_GEQ15min toCoupon_GEQ25min direction_same direction_opp Y

Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
temperature	12610	63.26725	19.15339	797800	30.00000	80.00000
age	10285	29.86874	7.68467	307200	21.00000	46.00000
has_children	12610	0.41451	0.49266	5227	0	1.00000
toCoupon_GEQ5min	12610	1.00000	0	12610	1.00000	1.00000
toCoupon_GEQ15min	12610	0.55979	0.49643	7059	0	1.00000
toCoupon_GEQ25min	12610	0.11602	0.32026	1463	0	1.00000
direction_same	12610	0.21554	0.41121	2718	0	1.00000
direction_opp	12610	0.78446	0.41121	9892	0	1.00000
Y	12610	0.56757	0.49543	7157	0	1.00000

Pearson Correlation Coefficients
Prob > |r| under H0: Rho=0
Number of Observations

	temperature	age	has_children	toCoupon_GEQ5min	toCoupon_GEQ15min	toCoupon_GEQ25min	direction_same	direction_opp	Y
temperature	1.00000 12610	-0.02094 0.0337 10285	-0.01860 0.0367 12610	. . 12610	-0.15709 <.0001 12610	-0.22717 <.0001 12610	0.09797 <.0001 12610	-0.09797 <.0001 12610	0.05939 <.0001 12610
age	-0.02094 0.0337 10285	1.00000 10285	0.42252 <.0001 10285	. . 10285	0.04012 <.0001 10285	-0.00691 0.4833 10285	-0.02599 0.0084 10285	0.02599 0.0084 10285	-0.03005 0.0023 10285
has_children	-0.01860 0.0367 12610	0.42252 <.0001 10285	1.00000 12610	. . 12610	0.07943 <.0001 12610	-0.01077 0.2264 12610	-0.03235 0.0003 12610	0.03235 0.0003 12610	-0.04506 <.0001 12610
toCoupon_GEQ5min	. . 12610	. . 10285	. . 12610	. . 12610	. . 12610	. . 12610	. . 12610	. . 12610	. . 12610
toCoupon_GEQ15min	-0.15709 <.0001 12610	0.04012 <.0001 10285	0.07943 <.0001 12610	. . 12610	1.00000 12610	0.32126 <.0001 12610	-0.30207 <.0001 12610	0.30207 <.0001 12610	-0.08269 <.0001 12610
toCoupon_GEQ25min	-0.22717 <.0001 12610	-0.00691 0.4833 10285	-0.01077 0.2264 12610	. . 12610	0.32126 <.0001 12610	1.00000 12610	-0.18990 <.0001 12610	0.18990 <.0001 12610	-0.10814 <.0001 12610

Pearson Correlation Coefficients
Prob > |r| under H0: Rho=0
Number of Observations

	temperature	age	has_children	toCoupon_GEQ5min	toCoupon_GEQ15min	toCoupon_GEQ25min	direction_same	direction_opp	Y
direction_same	0.09797 <.0001 12610	-0.02599 0.0084 10285	-0.03235 0.0003 12610	. . 12610	-0.30207 <.0001 12610	-0.18990 <.0001 12610	1.00000 12610	-1.00000 <.0001 12610	0.01493 0.0936 12610
direction_opp	-0.09797 <.0001 12610	0.02599 0.0084 10285	0.03235 0.0003 12610	. . 12610	0.30207 <.0001 12610	0.18990 <.0001 12610	-1.00000 <.0001 12610	1.00000 12610	-0.01493 0.0936 12610
Y	0.05939 <.0001 12610	-0.03005 0.0023 10285	-0.04506 <.0001 12610	. . 12610	-0.08269 <.0001 12610	-0.10814 <.0001 12610	0.01493 0.0936 12610	-0.01493 0.0936 12610	1.00000 12610

Univariate Analysis of Numeric Variables

The UNIVARIATE Procedure
Variable: temperature

Moments			
N	12610	**Sum Weights**	12610
Mean	63.2672482	**Sum Observations**	797800
Std Deviation	19.1533861	**Variance**	366.852199
Skewness	-0.6400003	**Kurtosis**	-1.0264086
Uncorrected SS	55100250	**Corrected SS**	4625639.37
Coeff Variation	30.2737777	**Std Error Mean**	0.17056425

Basic Statistical Measures			
Location		**Variability**	
Mean	63.26725	**Std Deviation**	19.15339
Median	80.00000	**Variance**	366.85220
Mode	80.00000	**Range**	50.00000
		Interquartile Range	25.00000

Tests for Location: Mu0=0				
Test	**Statistic**		**p Value**	
Student's t	**t**	370.9291	**Pr > |t|**	<.0001
Sign	**M**	6305	**Pr >= |M|**	<.0001
Signed Rank	**S**	39756178	**Pr >= |S|**	<.0001

Tests for Normality				
Test	**Statistic**		**p Value**	
Kolmogorov-Smirnov	**D**	0.322318	**Pr > D**	<0.0100
Cramer-von Mises	**W-Sq**	211.792	**Pr > W-Sq**	<0.0050
Anderson-Darling	**A-Sq**	1336.069	**Pr > A-Sq**	<0.0050

Quantiles (Definition 5)	
Level	**Quantile**
100% Max	80
99%	80
95%	80
90%	80
75% Q3	80
50% Median	80
25% Q1	55
10%	30
5%	30
1%	30
0% Min	30

Extreme Observations			
Lowest		**Highest**	
Value	**Obs**	**Value**	**Obs**
30	10541	80	12606

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
30	10540	80	12607
30	10539	80	12608
30	10538	80	12609
30	10537	80	12610

Univariate Analysis of Numeric Variables

The UNIVARIATE Procedure
Variable: age

Moments			
N	10285	Sum Weights	10285
Mean	29.8687409	Sum Observations	307200
Std Deviation	7.68466931	Variance	59.0541424
Skewness	0.56371587	Kurtosis	-0.7447647
Uncorrected SS	9782990	Corrected SS	607312.8
Coeff Variation	25.7281328	Std Error Mean	0.07577449

Basic Statistical Measures			
Location		Variability	
Mean	29.86874	Std Deviation	7.68467
Median	26.00000	Variance	59.05414
Mode	21.00000	Range	25.00000
		Interquartile Range	15.00000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	394.1794	Pr > |t|	<.0001
Sign	M	5142.5	Pr >= |M|	<.0001
Signed Rank	S	26447878	Pr >= |S|	<.0001

Tests for Normality				
Test	Statistic		p Value	
Kolmogorov-Smirnov	D	0.19729	Pr > D	<0.0100
Cramer-von Mises	W-Sq	57.79025	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	375.4919	Pr > A-Sq	<0.0050

Quantiles (Definition 5)	
Level	Quantile
100% Max	46
99%	46
95%	46
90%	41
75% Q3	36
50% Median	26
25% Q1	21
10%	21
5%	21
1%	21
0% Min	21

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
21	12609	46	12533
21	12608	46	12534
21	12607	46	12603
21	12559	46	12604

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
21	12558	46	12605

Missing Values			
Missing Value	Count	Percent Of	
		All Obs	Missing Obs
.	2325	18.44	100.00

Average Acceptance Rate by Coupon Type

The MEANS Procedure

Analysis Variable : Y			
coupon	N Obs	Mean	N
Bar	2010	0.4099502	2010
Carry out & Take away	2344	0.7337884	2344
Coffee House	3989	0.4986212	3989
Restaurant(20-50)	1488	0.4415323	1488
Restaurant(<20)	2779	0.7078086	2779

Acceptance Rate by Passenger Type and Coupon Type

The MEANS Procedure

Analysis Variable : Y				
passanger	coupon	N Obs	Mean	N
Alone	Bar	1198	0.4073456	1198
	Carry out & Take away	1361	0.7252021	1361
	Coffee House	2253	0.4376387	2253
	Restaurant(20-50)	1006	0.4234592	1006
	Restaurant(<20)	1430	0.6391608	1430
Friend(s)	Bar	335	0.5582090	335
	Carry out & Take away	737	0.7571235	737
	Coffee House	1226	0.5962480	1226
	Restaurant(20-50)	168	0.4642857	168
	Restaurant(<20)	826	0.8014528	826
Kid(s)	Bar	204	0.2107843	204
	Carry out & Take away	151	0.7019868	151
	Coffee House	206	0.4805825	206
	Restaurant(20-50)	173	0.3699422	173
	Restaurant(<20)	267	0.7228464	267
Partner	Bar	273	0.3882784	273
	Carry out & Take away	95	0.7263158	95
	Coffee House	304	0.5690789	304
	Restaurant(20-50)	141	0.6312057	141
	Restaurant(<20)	256	0.7734375	256

Acceptance Rate by Expiration and Coupon Type

The MEANS Procedure

Analysis Variable : Y				
expiration	coupon	N Obs	Mean	N
1d	Bar	1553	0.4121056	1553
	Carry out & Take away	1411	0.7810064	1411
	Coffee House	1763	0.5830970	1763
	Restaurant(20-50)	953	0.5215110	953
	Restaurant(<20)	1351	0.8312361	1351

Analysis Variable : Y				
expiration	coupon	N Obs	Mean	N
2h	Bar	457	0.4026258	457
	Carry out & Take away	933	0.6623794	933
	Coffee House	2226	0.4317161	2226
	Restaurant(20-50)	535	0.2990654	535
	Restaurant(<20)	1428	0.5910364	1428

Average Acceptance Rate by Weather x Time of Day

The MEANS Procedure

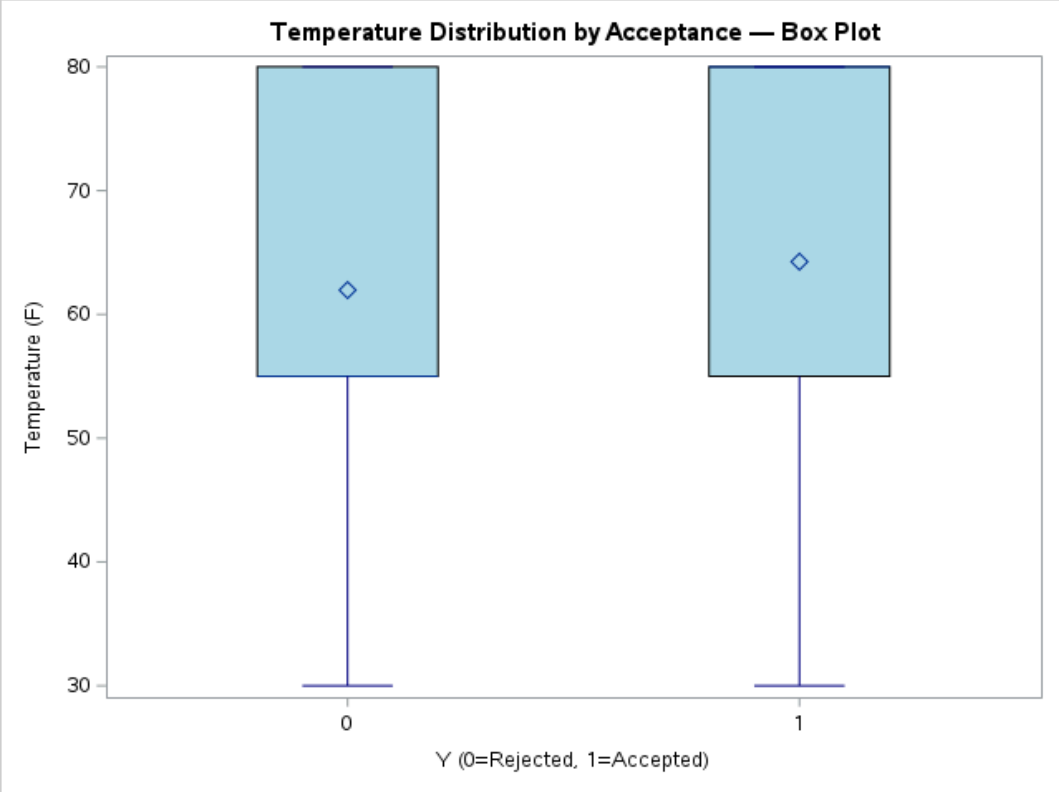
Analysis Variable : Y				
weather	time	N Obs	Mean	N
Rainy	10AM	214	0.5607477	214
	10PM	275	0.3709091	275
	2PM	67	0.6417910	67
	6PM	223	0.5560538	223
	7AM	423	0.3971631	423
Snowy	10AM	151	0.7483444	151
	10PM	390	0.3692308	390
	2PM	211	0.5355450	211
	6PM	159	0.3396226	159
	7AM	486	0.4835391	486
Sunny	10AM	1906	0.6017838	1906
	10PM	1334	0.5772114	1334
	2PM	1728	0.6776620	1728
	6PM	2838	0.6000705	2838
	7AM	2205	0.5215420	2205

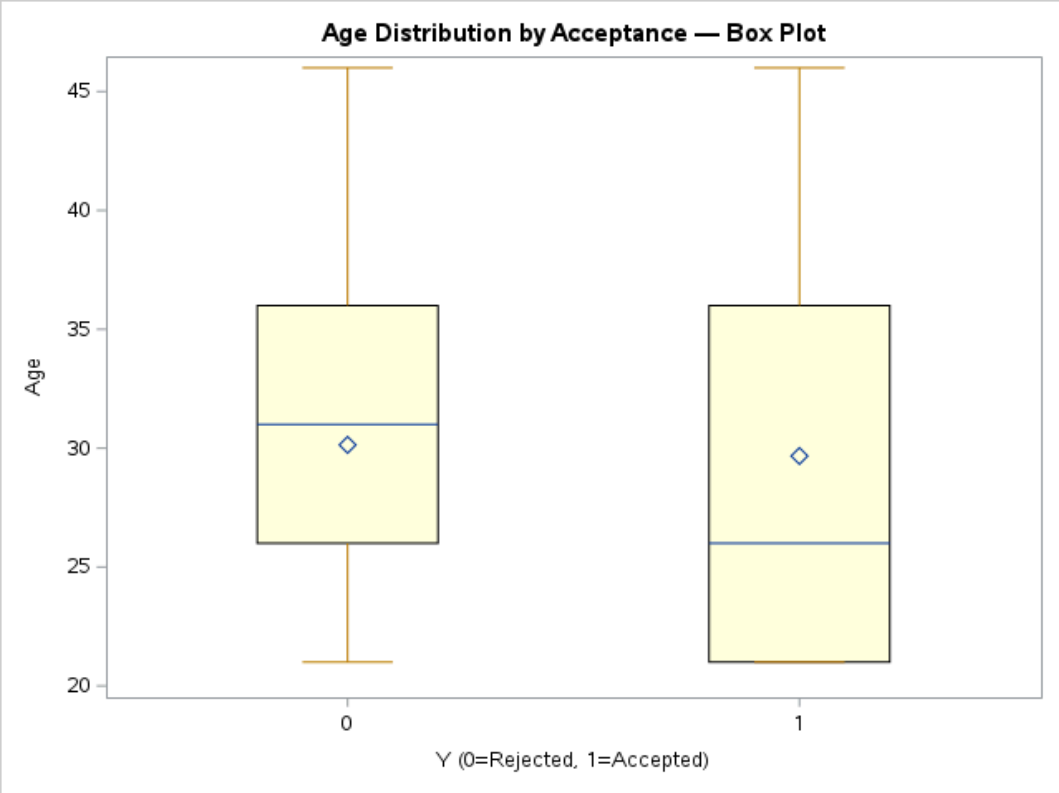
Outlier Count: IQR Method (1 = Outlier)

The FREQ Procedure

temp_outlier	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	12610	100.00	12610	100.00

age_outlier	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	10285	81.56	10285	81.56
1	2325	18.44	12610	100.00





1. Initial Missing Values Report (Numeric Variables)

The MEANS Procedure

Variable	N	N Miss	Minimum	Maximum
age	10285	2325	21.0000000	46.0000000
temperature	12610	0	30.0000000	80.0000000

2. Missing Percentage Across All Variables

The FREQ Procedure

destination	Frequency	Percent
Not Missing	12610	100.00

passanger	Frequency	Percent
Not Missing	12610	100.00

weather	Frequency	Percent
Not Missing	12610	100.00

temperature	Frequency	Percent
Not Missing	12610	100.00

time	Frequency	Percent
Not Missing	12610	100.00

coupon	Frequency	Percent
Not Missing	12610	100.00

expiration	Frequency	Percent
Not Missing	12610	100.00

gender	Frequency	Percent
Not Missing	12610	100.00

age	Frequency	Percent
Missing	2325	18.44
Not Missing	10285	81.56

maritalStatus	Frequency	Percent
Not Missing	12610	100.00

has_children	Frequency	Percent
Not Missing	12610	100.00

education	Frequency	Percent
Not Missing	12610	100.00

occupation	Frequency	Percent
Not Missing	12610	100.00

income	Frequency	Percent
Not Missing	12610	100.00

car	Frequency	Percent
Missing	12502	99.14
Not Missing	108	0.86

Bar	Frequency	Percent
Missing	107	0.85

Bar	Frequency	Percent
Not Missing	12503	99.15

CoffeeHouse	Frequency	Percent
Missing	217	1.72
Not Missing	12393	98.28

CarryAway	Frequency	Percent
Missing	150	1.19
Not Missing	12460	98.81

RestaurantLessThan20	Frequency	Percent
Missing	129	1.02
Not Missing	12481	98.98

Restaurant20To50	Frequency	Percent
Missing	189	1.50
Not Missing	12421	98.50

toCoupon_GEQ5min	Frequency	Percent
Not Missing	12610	100.00

toCoupon_GEQ15min	Frequency	Percent
Not Missing	12610	100.00

toCoupon_GEQ25min	Frequency	Percent
Not Missing	12610	100.00

direction_same	Frequency	Percent
Not Missing	12610	100.00

direction_opp	Frequency	Percent
Not Missing	12610	100.00

Y	Frequency	Percent
Not Missing	12610	100.00

3. Train/Test Split (80/20)

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling

Input Data Set	PREP
Random Number Seed	12345
Sampling Rate	0.8
Sample Size	10088
Selection Probability	0.8
Sampling Weight	0
Output Data Set	SPLIT_RESULTS

3. Train/Test Split (80/20)

The FREQ Procedure

Selection Indicator				
Selected	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	2522	20.00	2522	20.00
1	10088	80.00	12610	100.00

4. Final Validation - Missing Values (After Cleaning)

The MEANS Procedure

Analysis Variable : age	
N Miss	N
0	10088

4. Final Validation - Missing Values (After Cleaning)

The MEANS Procedure

Analysis Variable : age	
N Miss	N
0	2522

5. Final Frequency Distribution (Train Set)

The FREQ Procedure

Bar	Frequency	Percent
1 to 3 times	1941	19.24
4 to 8 times	842	8.35
Less than once	2758	27.34
More than 8 times	261	2.59
Never	4286	42.49

CoffeeHouse	Frequency	Percent
1 to 3 times	2562	25.40
4 to 8 times	1413	14.01
Less than once	2855	28.30
More than 8 times	881	8.73
Never	2377	23.56

CarryAway	Frequency	Percent
1 to 3 times	3812	37.79
4 to 8 times	3400	33.70
Less than once	1487	14.74
More than 8 times	1263	12.52
Never	126	1.25

RestaurantLessThan20	Frequency	Percent
1 to 3 times	4442	44.03
4 to 8 times	2800	27.76
Less than once	1654	16.40
More than 8 times	1024	10.15
Never	168	1.67

Restaurant20To50	Frequency	Percent
1 to 3 times	2572	25.50
4 to 8 times	586	5.81
Less than once	5012	49.68

Restaurant20To50	Frequency	Percent
More than 8 times	213	2.11
Never	1705	16.90

6. Final Frequency Distribution (Test Set)

The FREQ Procedure

Bar	Frequency	Percent
1 to 3 times	527	20.90
4 to 8 times	229	9.08
Less than once	680	26.96
More than 8 times	87	3.45
Never	999	39.61

CoffeeHouse	Frequency	Percent
1 to 3 times	637	25.26
4 to 8 times	366	14.51
Less than once	724	28.71
More than 8 times	226	8.96
Never	569	22.56

CarryAway	Frequency	Percent
1 to 3 times	983	38.98
4 to 8 times	842	33.39
Less than once	362	14.35
More than 8 times	309	12.25
Never	26	1.03

RestaurantLessThan20	Frequency	Percent
1 to 3 times	1043	41.36
4 to 8 times	753	29.86
Less than once	417	16.53
More than 8 times	258	10.23
Never	51	2.02

Restaurant20To50	Frequency	Percent
1 to 3 times	694	27.52
4 to 8 times	142	5.63
Less than once	1218	48.30
More than 8 times	51	2.02
Never	417	16.53

Final Standardized TRAIN Dataset Ready for Modeling

Obs	age	temperature	Total_Distance_Level	Coupon_Habit_Match	Urgency_Score	Bar_num	CoffeeHouse_num
1	0.0131906414	-0.431923096	1	1	4	0	3
2	0.0131906414	-0.431923096	1	1	4	0	3
3	0.0131906414	-0.431923096	1	1	4	1	4
4	0.0131906414	-0.431923096	1	1	4	1	3
5	-0.562235049	-0.431923096	1	0	4	1	1
6	-0.562235049	-0.431923096	1	0	4	0	1
7	0.8763291764	-0.431923096	1	0	4	0	1
8	0.0131906414	-0.431923096	1	0	4	0	0
9	-1.281517161	-0.431923096	1	0	4	0	1
10	-1.281517161	-0.431923096	1	0	4	3	0
11	-1.281517161	-0.431923096	1	1	4	1	3

Obs	age	temperature	Total_Distance_Level	Coupon_Habit_Match	Urgency_Score	Bar_num	CoffeeHouse_num
12	-1.281517161	-0.431923096	1	1	4	1	2
13	-0.562235049	-0.431923096	1	1	4	3	2
14	-0.562235049	-0.431923096	1	1	4	4	4
15	-0.562235049	-0.431923096	1	0	4	3	1
16	0.1570470639	-0.431923096	1	0	4	3	0
17	0.1570470639	-0.431923096	1	0	4	2	1
18	0.8763291764	-0.431923096	1	0	4	0	0
19	2.3148934014	-0.431923096	1	1	4	0	4
20	0.0131906414	-0.431923096	1	0	3	0	1

7. Logistic Regression Model - Training Phase

The LOGISTIC Procedure

Model Information	
Data Set	WORK.TRAIN_READY
Response Variable	Y
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	10088
Number of Observations Used	10088

Response Profile		
Ordered Value	Y	Total Frequency
1	0	4393
2	1	5695

Probability modeled is Y='1'.

Stepwise Selection Procedure

Step 0. Intercept entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

-2 Log L	=	13816.426

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
1230.3037	8	<.0001

Step 1. Effect Coupon_Habit_Match entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	13818.426	12922.702
SC	13825.645	12937.140
-2 Log L	13816.426	12918.702

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	897.7247	1	<.0001
Score	889.4732	1	<.0001
Wald	859.5773	1	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
374.5674	7	<.0001

Note: No effects for the model in Step 1 are removed.

Step 2. Effect Urgency_Score entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	13818.426	12617.918
SC	13825.645	12639.575
-2 Log L	13816.426	12611.918

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	1204.5083	2	<.0001
Score	1168.1358	2	<.0001
Wald	1076.9684	2	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
69.3969	6	<.0001

Note: No effects for the model in Step 2 are removed.

Step 3. Effect temperature entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	13818.426	12578.830
SC	13825.645	12607.706
-2 Log L	13816.426	12570.830

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	1245.5967	3	<.0001
Score	1204.9191	3	<.0001
Wald	1105.6020	3	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
28.3056	5	<.0001

Note: No effects for the model in Step 3 are removed.

Step 4. Effect CarryAway_num entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	13818.426	12566.624
SC	13825.645	12602.719
-2 Log L	13816.426	12556.624

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	1259.8022	4	<.0001
Score	1217.3622	4	<.0001
Wald	1114.9229	4	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
14.1559	4	0.0068

Note: No effects for the model in Step 4 are removed.

Step 5. Effect Total_Distance_Level entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	13818.426	12557.299
SC	13825.645	12600.614
-2 Log L	13816.426	12545.299

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	1271.1268	5	<.0001
Score	1227.7748	5	<.0001
Wald	1123.5246	5	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
2.8142	3	0.4212

Note: No effects for the model in Step 5 are removed.

Note: No (additional) effects met the 0.05 significance level for entry into the model.

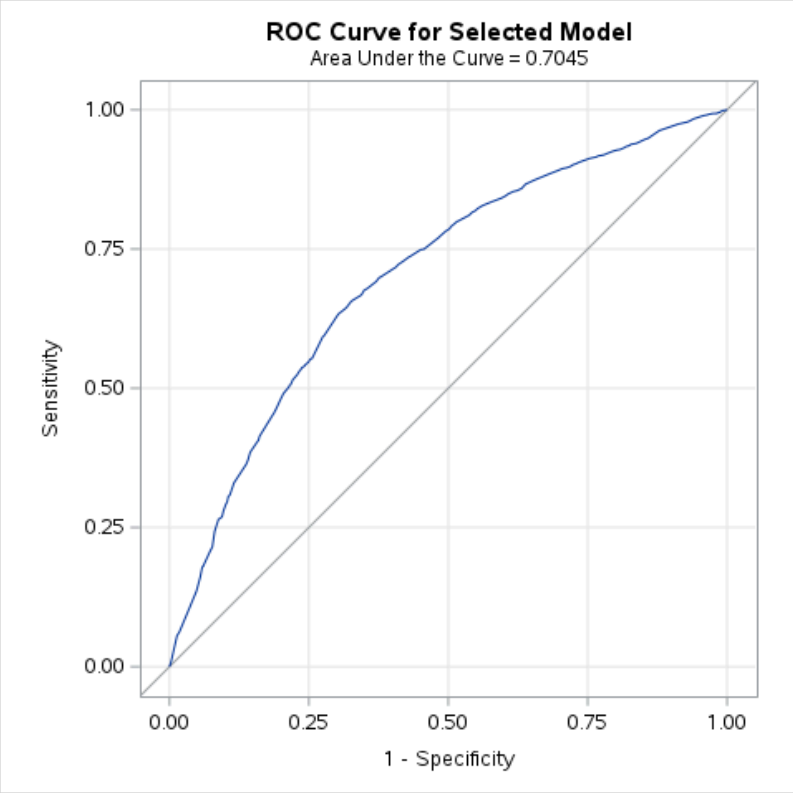
Summary of Stepwise Selection							
Step	Effect Entered	Effect Removed	DF	Number In	Score Chi-Square	Wald Chi-Square	Pr > ChiSq
1	Coupon_Habit_Match		1	1	889.4732		<.0001
2	Urgency_Score		1	2	305.9982		<.0001
3	temperature		1	3	41.1717		<.0001
4	CarryAway_num		1	4	14.1870		0.0002
5	Total_Distance_Level		1	5	11.3523		0.0008

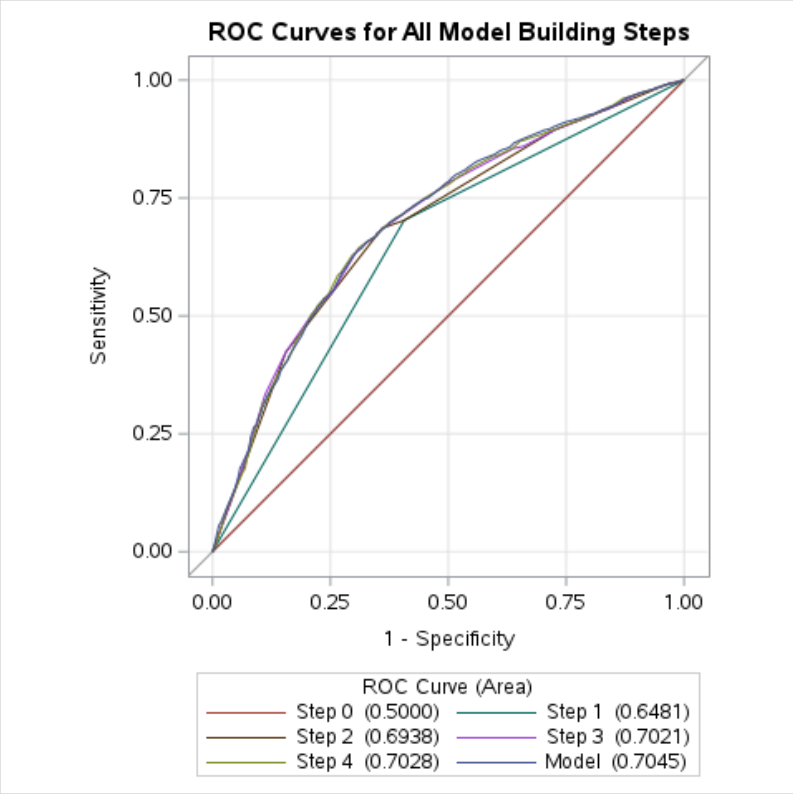
Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-1.2983	0.1286	101.9841	<.0001
temperature	1	0.1202	0.0221	29.5041	<.0001
Total_Distance_Level	1	-0.1167	0.0346	11.3405	0.0008
Urgency_Score	1	0.3754	0.0243	239.0721	<.0001
Coupon_Habit_Match	1	1.3517	0.0451	896.6121	<.0001
CarryAway_num	1	-0.0884	0.0237	13.9250	0.0002

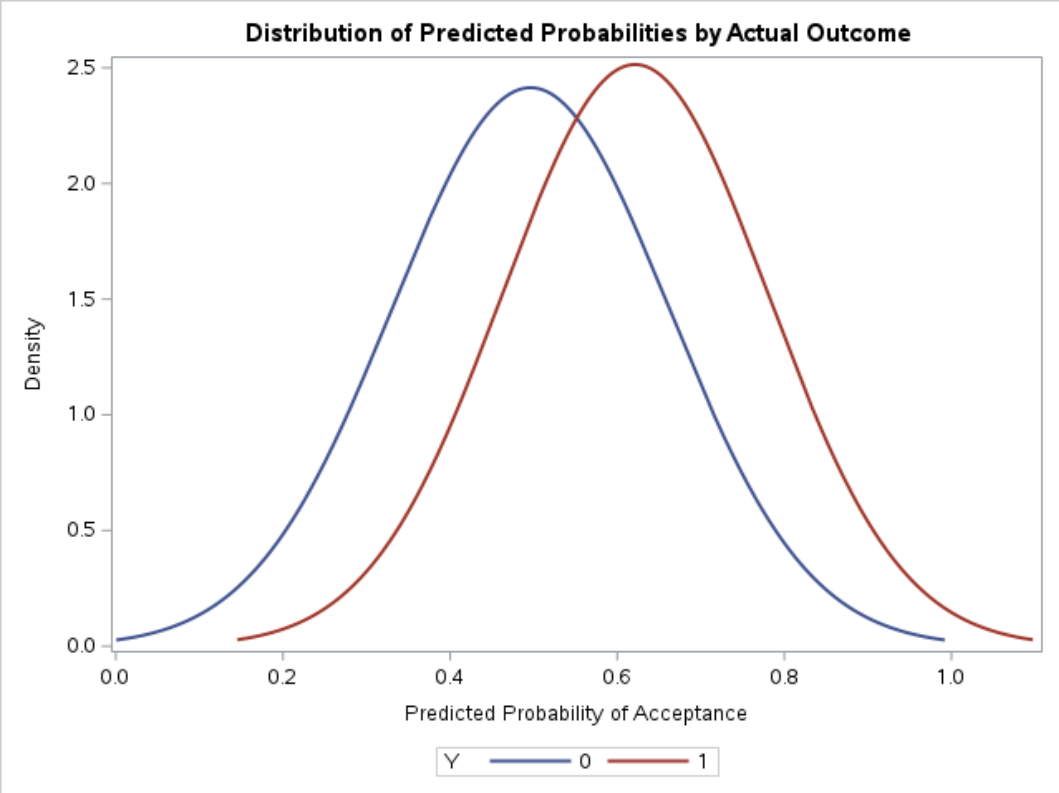
Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
temperature	1.128	1.080	1.178
Total_Distance_Level	0.890	0.831	0.952
Urgency_Score	1.456	1.388	1.526

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Coupon_Habit_Match	3.864	3.537	4.222
CarryAway_num	0.915	0.874	0.959

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	69.9	Somers' D	0.409
Percent Discordant	29.0	Gamma	0.414
Percent Tied	1.2	Tau-a	0.201
Pairs	25018135	c	0.704







Confusion Matrix (Test Set)

The FREQ Procedure

Frequency	Table of Y by I_Y		
	I_Y(Into: Y)		
Y	0	1	Total
0	628	432	1060

Table of Y by I_Y			
	I_Y(Into: Y)		
Y	0	1	Total
1	385	1077	1462
Total	1013	1509	2522

Confusion Matrix (Test Set)

The DISCRIM Procedure

Total Sample Size	10088	DF Total	10087
Variables	8	DF Within Classes	10086
Classes	2	DF Between Classes	1

Number of Observations Read	10088
Number of Observations Used	10088

Class Level Information					
Y	Variable Name	Frequency	Weight	Proportion	Prior Probability
0	0	4393	4393	0.435468	0.500000
1	1	5695	5695	0.564532	0.500000

Confusion Matrix (Test Set)

The DISCRIM Procedure
Classification Summary for Calibration Data: WORK.TRAIN_READY
Resubstitution Summary using 5 Nearest Neighbors

Number of Observations and Percent Classified into Y			
From Y	0	1	Total
0	3178 72.34	1215 27.66	4393 100.00
1	1437 25.23	4258 74.77	5695 100.00
Total	4615 45.75	5473 54.25	10088 100.00
Priors	0.5	0.5	

Error Count Estimates for Y			
	0	1	Total
Rate	0.2766	0.2523	0.2645
Priors	0.5000	0.5000	

Confusion Matrix (Test Set)

The DISCRIM Procedure
Classification Summary for Test Data: WORK.TEST_READY
Classification Summary using 5 Nearest Neighbors

Observation Profile for Test Data	
Number of Observations Read	2522
Number of Observations Used	2522

Number of Observations and Percent Classified into Y			
From Y	0	1	Total
0	653 61.60	407 38.40	1060 100.00
1	478 32.69	984 67.31	1462 100.00
Total	1131 44.85	1391 55.15	2522 100.00

Number of Observations and Percent Classified into Y			
From Y	0	1	Total
Priors	0.5	0.5	

Error Count Estimates for Y			
	0	1	Total
Rate	0.3840	0.3269	0.3555
Priors	0.5000	0.5000	

Performance of KNN Model (K=5)

The FREQ Procedure

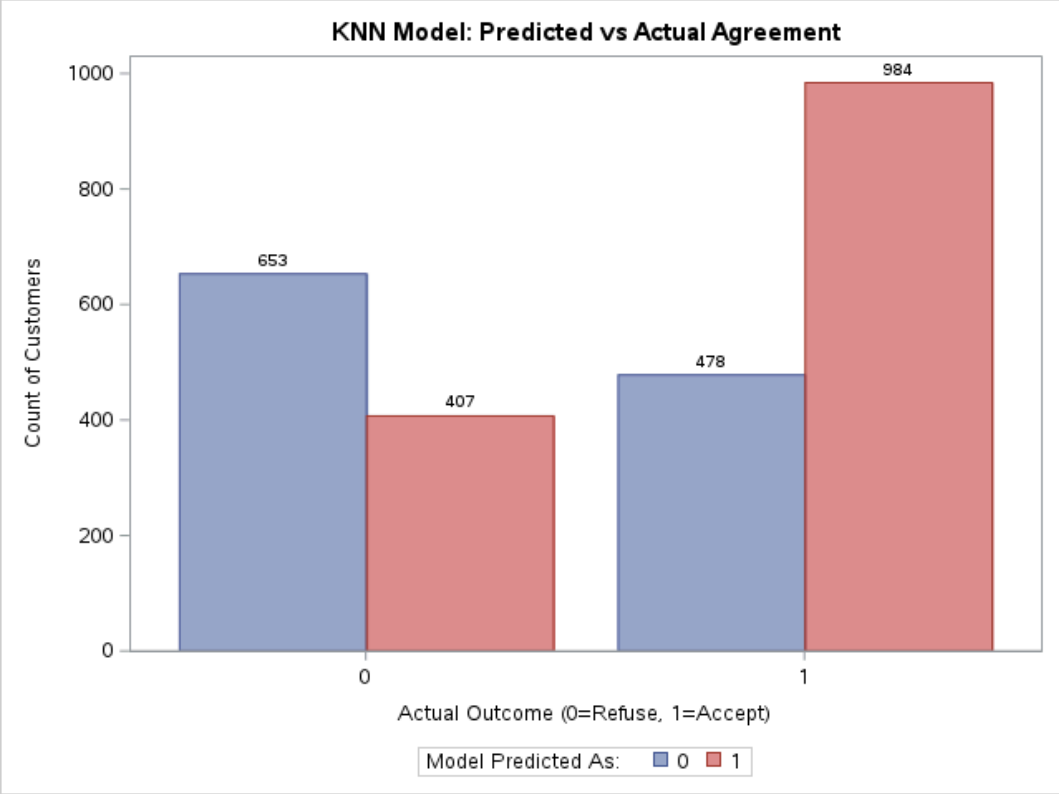
Frequency Percent Row Pct Col Pct

Table of Y by _INTO_			
	INTO		
Y	0	1	Total
0	653 25.89 61.60 57.74	407 16.14 38.40 29.26	1060 42.03
1	478 18.95 32.69 42.26	984 39.02 67.31 70.74	1462 57.97
Total	1131 44.85	1391 55.15	2522 100.00

Statistics for Table of Y by _INTO_

Sensitivity and Specificity				
Statistic	Estimate	Standard Error	95% Confidence Limits	
Sensitivity	0.5774	0.0147	0.5486	0.6062
Specificity	0.7074	0.0122	0.6835	0.7313
Positive Predictive Value	0.6160	0.0149	0.5868	0.6453
Negative Predictive Value	0.6731	0.0123	0.6490	0.6971

Sample Size = 2522


KNN Model: Predicted vs Actual Agreement
1000
800
600
400
200
0
Count of Customers
653
407
478
984
0
1
Actual Outcome (0=Refuse, 1=Accept)
Model Predicted As: 0 1